Printers and scanners

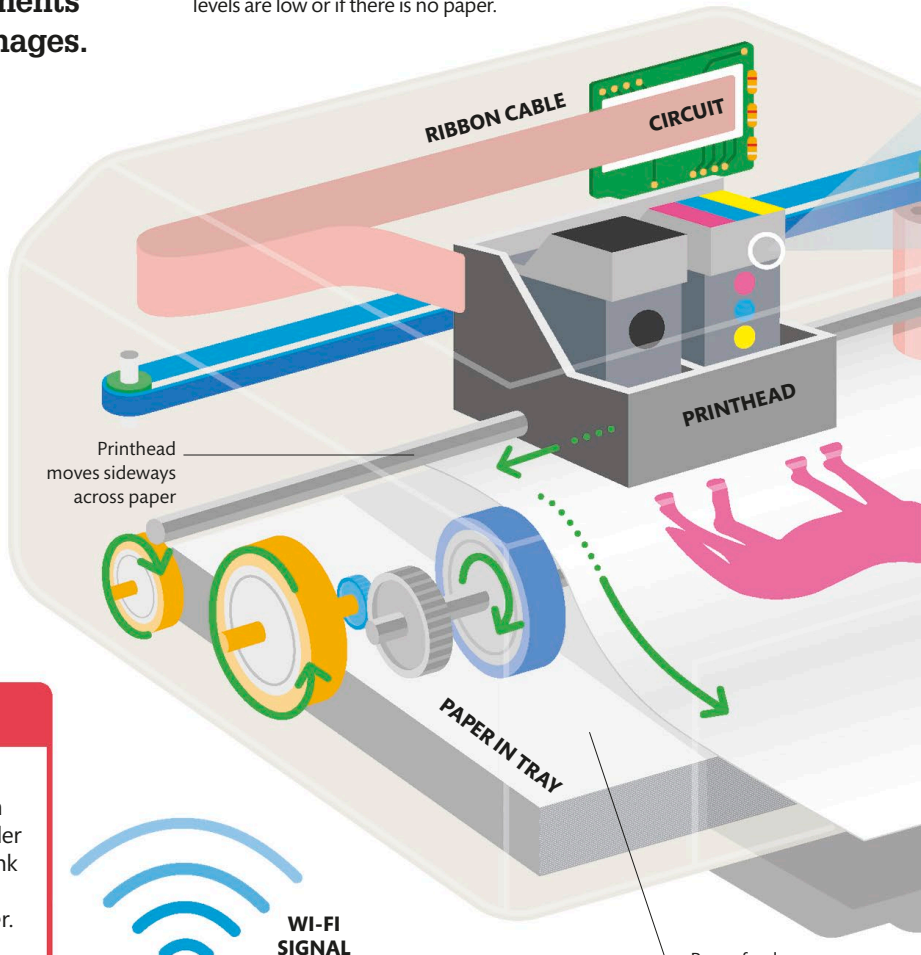
Computer printers enable us to output documents and photographs stored on a computer or other digital device, while scanners capture documents and photographs as digital images.

Inkjet printers

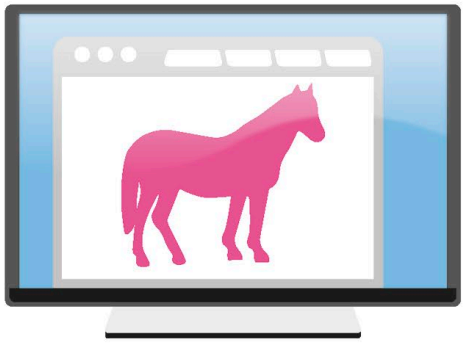
The most common type of printer uses a jet of ink droplets to form images and text on the printed page. In the printer, ink cartridges move back and forth, spraying ink onto the paper as it advances beneath them. A color image is made of millions of dots of four colored inks: yellow, magenta, cyan, and black. In many printers, the three nonblack inks are held in one cartridge. Each color is delivered individually, and they combine to give subtle variations in hue and tone. The cartridge heads have hundreds of holes through which the ink is forced.

Printer receives data from computer via Wi-Fi

2 Message received by printer
Software inside the printer processes the document or image, taking account of the desired paper size. It also communicates back to the computer if ink levels are low or if there is no paper.



WI-FI SIGNAL



1 Image sent to printer
The computer prepares the image or document, representing it as a collection of binary digits (see p.158) that the printer can process and sends it through a cable or across a wireless network.

LASER PRINTERS

A laser scans across a rotating drum, which develops a negative charge where the beam hits. Positively charged toner sticks to the roller where the laser has hit. The roller transfers ink to the paper, which then passes through heated rollers, fusing the toner to the paper.

